

architecture uses open APIs to enable quick and transparent integration with other enterprise applications and systems implemented by various organisations, thereby providing access to data and services, and promoting citizen/developer participation for the benefit of the community.

API management includes the full API management life cycle, integration with diverse applications and appliances to support management, and monitoring and tracking of all your API activities.

Large complex systems in an enterprise follow the API-centric approach for exposing services for integration to multiple third-party solutions. The APIs are RESTful, XML based, and stateless services. All APIs are accessible via the HTTPS protocol.

Service discoverability: This is the process of a client application querying the central registry to learn the location of services. Service discoverability ensures that metadata attached to a service describes its overall purpose and functionality, which makes the services easily discoverable. A repository of re-usable business logic components is maintained and made available as part of service discoverability.

Consul, Apache Zookeeper, Etc, Netflix Eureka, etc, are the largely used open source service discovery tools.

Platform and database-agnostic: Applications need to be forward and backward compatible, and be deployable on any technology platform. They should be able to communicate with any data store.

For example, Jenkins is a continuous delivery tool that builds and tests software after every change. Jenkins can deploy new code to production, which helps in no downtime in making the upgraded product available to constituents.

PostgreSQL, MySQL, MongoDB, CouchBase are the popular open source databases.

Microservices: The design of the business and information services

layer of the enterprise application is developed using ‘microservices’ architecture principles and cross-channel capabilities. Microservice architecture (MSA) allows the creation of services that are loosely coupled and have different programming language bases, are scalable, have a quicker delivery time, etc. Businesses can develop new enterprise applications as reusable microservices based on open standards. This will ensure independent scaling of individual services within an application with better resource utilisation.

Application life cycle automation tools like Ansible help in tracking, deploying and measuring the changes/enhancements of microservices.

Mule, Jboss, AngularJS, Bootstrap are the frequently used open source dev tools today. Other popular open source tools are Joomla, Jetty, OpenSSO, SOLR, Chef, Spring, and WordPress.

Integration: The value of a strong integration system lies in its ability to connect to a wide range of backend systems. It provides mediation, transformation, protocol and routing capabilities, and acts as a gateway to integrate with core backend systems. It also provides aggregation and broker communications. Therefore, it is important to consider an integration solution vendor that has the ability to connect in a predictable manner to other databases, frameworks, applications and endpoints.

It is important to go with open standards and open interfaces while choosing the integration backend, as it should be free from vendor-specific modifications and proprietary hooks.

Mule, JBoSS, and WS02 are popular open source integration products.

Security: Effective enterprise security is the need of the hour. It ensures that all the components of the cloud infrastructure are guarded against the fast growing security threats. Applications and endpoints need to be protected and users granted access to applications as per the eligibility

configured as part of the RBAC (role based access control).

Oauth2 and OpenID are open source tools that are used very often for this.

Customer experience: A good customer experience means that the individual’s experience during all points of contact with an application matches his or her expectations.

HTML5 and Joomla are a few open source tools used for enhancing the customer experience.

Analytics: Enterprises may apply analytics to better their customer services. Predictive analytics helps business services to be data driven and take decisions appropriately.

scikit-learn and TensorFlow are important open source tools in this domain.

PaaS: This is a category of cloud computing services that provides a platform allowing customers to develop, run, and manage applications without the complexity of building and maintaining the infrastructure typically associated with developing and launching an application. This is the core platform on which cloud native applications and supporting systems are based.

Cloud Foundry, Cloudify and OpenShift are a few popular open source PaaS platforms.

Key open source cloud platforms

When choosing an open source cloud platform, businesses must look at:

- Deployment flexibility
- Ease of operations
- Choice of application stacks
- Language, database and framework support
- Scaling capabilities
- Quality of service
- Tooling for development and operations
- Business fitment

A few important and popular open source cloud platforms are listed below.

OpenStack: OpenStack is open source software for creating private and